WASP-10b

R-1694

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_WASP-10_140921` · created 2026-08-02T13:31:03+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2456921.7271 +/- 0.0013 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1561 +/- 0.0074
Transit depth (Rp/Rs)^2	2.44 +/- 0.23 [%]
Orbital Inclination (inc)	89.24 +/- 0.58
Airmass coefficient 1 (a1)	0.994 +/- 0.01
Airmass coefficient 2 (a2)	0.007 +/- 0.014
Scatter in the residuals of the lightcurve fit is	1.03 %
Best Comparison Star	#2 - [286, 351]
Optimal Aperture	3.35
Optimal Annulus	4.63
Transit Duration (day)	0.0937 +/- 0.0016

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.1561 ± 0.0074 vs 0.1592 (deviation 0.42σ)
Duration fit vs published	0.0937 d vs 0.0946 d (deviation 0.55σ)
Mid-time O–C	-3.7 min
Depth SNR	21.1
Residual scatter	1.03 %

Light curve

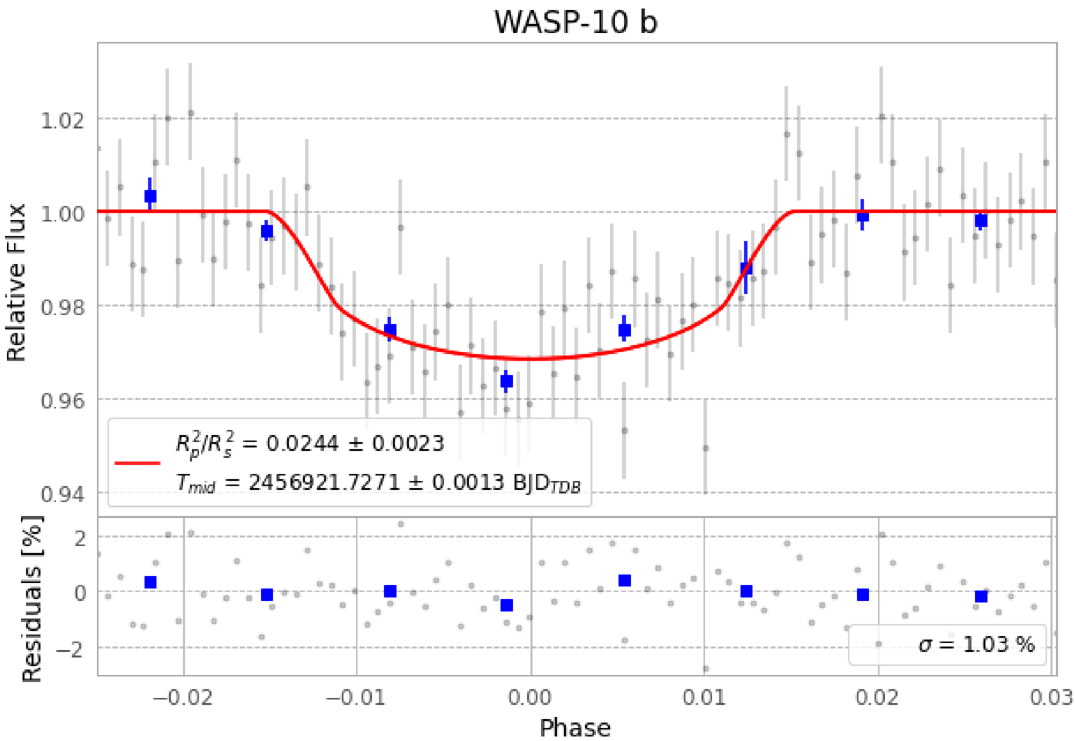

Figure 1 — FinalLightCurve_WASP-10 b_2014-09-20.png (SHA-256 b905941abc668a95…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [297, 226], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 10415bdb161fd03a…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b`

Data provenance

83 FITS frames (55 MB), `WASP-10140921032712.FITS` → `WASP-10140921073312.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-10-140921.log (3fd948a96a81…), FinalLightCurve_WASP-10 b_2014-09-20.png (b905941abc66…), FinalLightCurve_WASP-10 b_2014-09-20.csv (ea922020fb38…)

Compute resources

Wall time 160.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)